# INDIVIDUAL WORK ALLOCATION & DELIVERY PLAN

Multi-Dataset Outbreak Investigation & Evidence Reconstruction Workspace

## Anjali Kumari

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| Project | OUTBREAK REWIND |
| Planned workload share | 37.5% |
| Role | Frontend, Investigation Workspace & Visualisation Owner |
| Architecture baseline | Revised multi-dataset architecture — 12 November 2026 |
| Purpose | Individual ownership, tasks, deliverables, handoffs and acceptance criteria |

### Mission

Build the investigator-facing experience from dataset onboarding through analysis and evidence review, with clear visual communication of source, metric, uncertainty and data-quality limitations.

### Ownership boundary

Owns React/TypeScript application, navigation, design system, workspace, dataset and mapping screens, surveillance explorer, outbreak/event views, evidence/hypothesis/Rewind presentation, reports and frontend tests. Does not own backend persistence, file parsing or analytical calculations.

**Workload note: this role carries a larger share of implementation and integration responsibility. Keep interfaces explicit and raise scope changes early rather than silently absorbing another owner's module.**

## 1. Shared non-negotiable project rules

- Keep all original uploaded files unchanged and preserve dataset version, source-row identity and provenance.
- Do not flatten datasets with different grains or metric definitions into one table; retain dataset-specific extensions.
- Missing is not zero. Preserve observed, estimated, unknown and not-applicable distinctions.
- Do not infer person-level cases, contacts or transmission chains from aggregate or outbreak-level sources.
- Every metric and evidence item must be traceable to a dataset version and source record.
- Only compare sources when geography, period, metric definition and units are compatible; surface uncertainty.
- Evidence Rewind proposes deterministic changes and impact previews; a human must approve changes.
- Treat the product as an investigation/research workspace, not a clinical or operational public-health decision system.

## 2. Detailed task breakdown

### 1. Frontend foundation and navigation

- Set up React, TypeScript, Vite, routing, shared layout and reusable components.
- Build workspace navigation, investigation overview and responsive application shell.
- Create reusable tables, filters, detail panels, status chips, loading, empty, error and permission states.
- Consume documented API contracts rather than accessing database structures directly.

### 2. Dataset onboarding and investigation workspace

- Build dataset library, dataset detail, import progress, schema preview and field-mapping screens.
- Show original source-column names alongside mapped canonical fields; surface unmapped fields rather than hiding them.
- Present server-returned validation warnings, missingness, duplicate candidates and import status.
- Build investigation overview showing selected datasets, versions, date/geography coverage and unresolved warnings.

### 3. Surveillance and outbreak/event visualisation

- Build filters for disease/topic, geography, date range, source and observation status.
- Implement time-series and comparison charts only for compatible metrics and time grains returned by the API.
- Label units, reporting periods, dataset/source and observed-versus-estimated status on charts.
- Build outbreak/event register and detail screens using fields that are actually present in the source records.
- Do not create person-level case/contact screens or imply unsupported transmission networks.

### 4. Evidence, Rewind and report experience

- Build evidence list/detail screens with dataset version, source record and verification status.
- Build human-authored hypothesis workspace with supporting, contradicting and contextual evidence groupings.
- Build Rewind before/after preview, uncertain-match warnings, approve/reject/request-information actions and revision history.
- Build report preview/download screens from backend-provided report data; add responsive and accessibility basics.

## 3. Required deliverables

- Runnable React/TypeScript app with navigation and shared component system.
- Workspace, investigation overview, dataset library, dataset detail and field-mapping screens.
- Import/validation feedback screens and data-quality presentation.
- Surveillance explorer and outbreak/event register/detail screens.
- Evidence, hypothesis, Rewind review, revision history and report screens.
- Frontend tests, screenshots and UI handoff notes.

## 4. Definition of done and acceptance criteria

**A task is complete only when the implementation is committed, documented and supported by a reproducible test, working endpoint, migration or demonstrated workflow.**

- All agreed routes and primary controls work; no dead links or inert buttons.
- Charts show metric, unit, time grain and source; incompatible datasets are not silently combined.
- Missing, estimated, unknown and zero values are visually distinguishable when supplied by the API.
- Mapping and validation screens show source fields and server warnings without silently modifying records.
- Evidence details expose provenance; Rewind displays impact before a decision and resulting status afterward.
- Loading, empty, error and permission-denied states are implemented and tested.

## 5. Dependencies and handoffs

- Receive OpenAPI endpoints, filters, pagination, import states, mapping and validation payloads from Anjali Sagar.
- Receive evidence, hypothesis, rewind, review-decision and history contracts from Adithya.
- Provide API payload requirements, chart query needs, UI state definitions and end-to-end interaction tests to both owners.
- All three: agree route names, terminology, status colours and demo data before integration.

## 6. Milestones and sequencing

Sequence is dependency-based. The team should add calendar dates after confirming availability and the review deadline; no dates are invented in this plan.

| Milestone | Expected contribution |
|---|---|
| M0 | Confirm user journey, routes, screens and demo scenarios. |
| M1 | Deliver app shell, routing, shared components and API client. |
| M2 | Deliver dataset onboarding, mapping and validation screens. |
| M3 | Deliver surveillance and outbreak/event exploration. |
| M4 | Deliver evidence/Rewind/report screens and frontend tests. |

## 7. Explicitly out of scope

- FastAPI routes, database schema/migrations or backend authorization.
- CSV/ZIP parsing, schema inference or canonical normalization.
- Computing quality rules, statistical measures or cross-source conflict decisions in the browser.
- Inventing records, filling missing values or changing metric definitions in the UI.

## 8. Review/demo contribution

**Present the investigator journey from dataset selection to quality review, then a correctly labelled surveillance chart, an outbreak/event record and the Rewind review interface.**

## 9. Weekly progress update

- Completed: PR/commit, endpoint, migration, test result or document link.
- In progress: exact task, status and next action.
- Blocked: dependency, person who can unblock it and decision required.
- Evidence: reproducible command, test output or working demo.
- Risk/change: report scope, security or data-quality risks before implementation.

## 10. Collaboration and handoff checklist

- Use focused branches and pull requests; request review from affected module owners.
- Do not commit credentials, secrets or private datasets.
- Document setup and sample configuration without secrets.
- Keep field names, identifiers and status values aligned across all workstreams.
- Demonstrate from a clean checkout before marking a feature complete.

## 11. Shared project completion standard

- One aggregate-surveillance workflow and one outbreak/event workflow work end-to-end.
- Every displayed metric and evidence item can be traced to its dataset version and source record.
- Data limitations and unresolved warnings are visible.

- Rewind changes require human approval and preserve prior revisions.
- Known limitations are documented; mock-ups alone do not count as completed functionality.

Status note: this is a proposed work allocation derived from the revised architecture and supplied dataset inventory. It is not evidence that the listed features have been implemented, scientifically validated, load-tested or security-certified. Verify dataset permissions before deployment.